Ahmet Bera Çelik

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Summary

Final-year Computer Engineering student at Recep Tayyip Erdoğan University, working on AI, robotics and embedded systems. I lead the software team of our autonomous surface vehicle, a TEKNOFEST 2026 Mavi Vatan finalist, and serve as president of our department's student society. I like taking ownership in a team and backing my work with tests and documentation.

Education

Sept 2022 – June 2027 **Recep Tayyip Erdoğan University**, BSc in Computer Engineering – Rize, Türkiye

- Expected graduation June 2027. 30% English-medium program; completed the English preparatory year.
- Senior project: an edge-AI hardware and software system for IoT devices.

Experience

Sept 2025 – present **President**, RTEÜ Computer Engineering Society – Rize, Türkiye

- Run the society's management and event program. Previously served as Project Coordinator (2024–2025), planning workshops and handling participant communication.
- Taught a two-session workshop on Git & GitHub and Data Structures in November 2025.
- Organized a trip to two AI summits in Ankara with support from the Ministry of Youth and Sports, and prepared the society's grant application to the Ministry's ÜNİDES program.

July 2025 – Aug 2025 **Software Intern**, Teknopark İstanbul – İstanbul, Türkiye

- Completed a 30-working-day mandatory internship on site, taking part in software development work.

Projects

2026 – present **Autonomous Surface Vehicle (ZUGA-İDA), TEKNOFEST 2026 Mavi Vatan Finalist**

- Software team lead in a 15-member team of computer, electrical-electronics and mechanical engineering students.
- Built the vision and autonomous mission software with Python, YOLO and MAVLink; tested it in Gazebo and ArduPilot SITL simulation, then deployed it to a Jetson Orin Nano.
- Set up unit, integration and hardware-in-the-loop (HIL) tests; in the 12 August 2026 open-water test the vehicle completed buoy avoidance and the red-target task.

2026 **SecureBudget: Encrypted Personal Budget Book (Secure Programming course)**

- Captain of a five-person team building a C++ console application and a security library (OpenSSL, SQLite, SoftHSM, CMake, CTest).
- Set up the team workflow: protected main branch, pull requests with peer review, and GitHub Actions CI on Windows and Linux.

2026 **LifeLine: Offline Voice First-Aid Assistant (Internet of Things course)**

- Two-person project: an ESP32-S3 device that measures heart rate and speaks first-aid steps using local AI models, with no internet. Wrote the C firmware (ESP-IDF, FreeRTOS), the signal processing and the computer-side application.

2025 **Microservices CRM Platform**

- CRM with user, department and ticket services built on Java 17, Spring Boot, WebFlux, R2DBC, JWT and MSSQL; React and TypeScript frontend, deployed with Docker Compose.

2024 **SafeScript: Safe Prescribing Assistant for Physicians, TEKNOFEST 2024 Finalist**

- In the three-person team MedWise, built the frontend (JavaScript, Vite) of a web application that flags medications a patient should avoid.

Skills

- Programming languages: Python, C, C++, Java, JavaScript, TypeScript
- AI and robotics: YOLO, OpenCV, Gazebo, ArduPilot SITL, MAVLink, Jetson Orin Nano
- Web and embedded: Spring Boot, Node.js, React, Next.js, REST APIs, ESP32-S3, ESP-IDF, FreeRTOS
- Databases and tools: MySQL, MSSQL, SQLite, Git, GitHub Actions, Docker, CMake, CTest, JUnit 5, Linux
- AI-assisted software development: Claude Code, Cursor

Languages

- Turkish: native; English: intermediate (B1)